STAC PS2: SOLAR RADIATION ANALYSIS AND PREDICTION

Team Nexus

***Abstract*—This report analyses 32 686 five-minute observations of solar radiation, temperature, pressure, humidity and wind recorded by the Voyage-123 probe between 1 September and 31 December 2016. Exploratory visualisation shows a clean bell-shaped daily radiation profile whose peak envelope falls by roughly 21% from September to December, with a large cloud-driven scatter beneath it. Radiation peaks on average at 12:14:47 and the median peak time is 12:15:02. Temperature has the strongest linear relationship with radiation, whereas humidity is weakly and negatively related. A day-wise report built with Pandas and NumPy identifies 4 September 2016 as the day of maximum radiation (1601.26 W/m^2) and the five highest-average days, all in September. A random forest regressor trained on a 70/30 split reaches an RMSE of 91.45 W/m^2 and $R^2 = 0.915$; a chronological split, which is a stricter test, gives $R^2 = 0.526$. All plots are made in MatPlotLib**

***Index Terms*—solar radiation, exploratory data analysis, Pandas, MatPlotLib, random forest regression, time-series**

I. Introduction

Solar panels and communications links both depend on how much radiation reaches the surface, and that quantity changes with time of day, season and weather. The Voyage-123 probe logged radiation, temperature, pressure, humidity, wind speed and direction, and sunrise and sunset times. The task has four parts: (A) visualise the data, (B) draw inferences from the visualisations, (C) build a day-wise solar report, and a bonus part (D) that predicts radiation with a regression model. This report presents the plots, the inferences and the model scores for all four parts. All analysis uses Python with Pandas [**?**], NumPy [**?**], Matplotlib [**?**] and scikit-learn [**?**].

II. Dataset and Preprocessing

The dataset contains 32 686 records and 11 columns. Observations cover 118 consecutive days (1 September to 31 December 2016) at roughly five-minute intervals. Time of day was converted to decimal hours, h = seconds since midnight$/3600$, for plotting. Records were sorted by `UNIXTime`. Each day's group was re-ordered by its first timestamp.

Radiation has mean 207.1 W/m^2, standard deviation 315.9 W/m^2, a minimum of 1.11 W/m^2 and a maximum of 1601.26 W/m^2. The night-time floor of about 1.3 W/m^2 is a sensor offset. Temperature is recorded in °Fahrenheit (34–71°F), humidity in percent (8–103%) and pressure in inches of mercury (30.19–30.56).

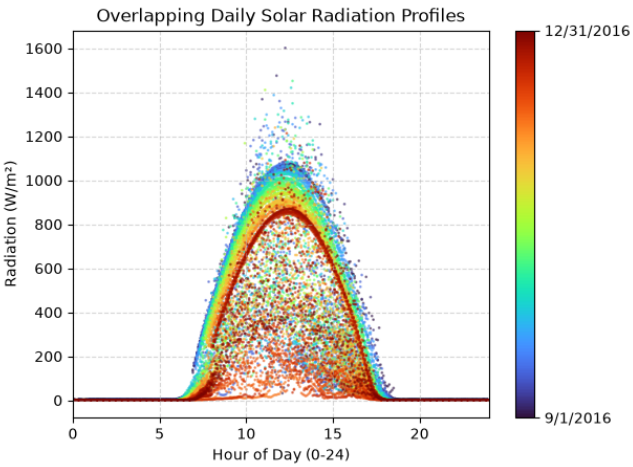


Fig. 1. Overlapping daily solar radiation profiles, coloured by date (1 September to 31 December 2016).

III. Part A: Data Visualisation

A. Radiation against time

Fig. 1 overlays all 118 daily radiation profiles against the hour of day, coloured by date. Days generally follow the same bell shape between about 06:00 and 18:00, and the profile is flat at the sensor floor overnight. The upper boundary is a smooth clear-sky envelope that lowers steadily from September (blue) to December (dark red). Below that envelope lies a dense cloud of points from cloudy intervals.

B. Radiation against temperature and humidity

Fig. 2 plots radiation against temperature and against humidity. High radiation occurs almost only at warmer temperatures, and the colour gradient shows that the warmest readings come from September while the coolest come from December. Humidity shows the opposite tendency: the densest cluster at high humidity and low radiation is December, and most readings above about 1000 W/m^2 sit at moderate humidity.

C. Bonus: daily pressure cycle

Fig. 3 subtracts each day's mean pressure and plots the deviation against hour of day. The mean over all days (black) traces a clear two-peak-per-day oscillation: maxima near 09:00–10:00 (about $+0.028$ inHg) and 21:00–22:00 ($+0.024$ inHg), and minima near 03:00–04:00 (-0.021 inHg) and 15:00 (-0.027 inHg).

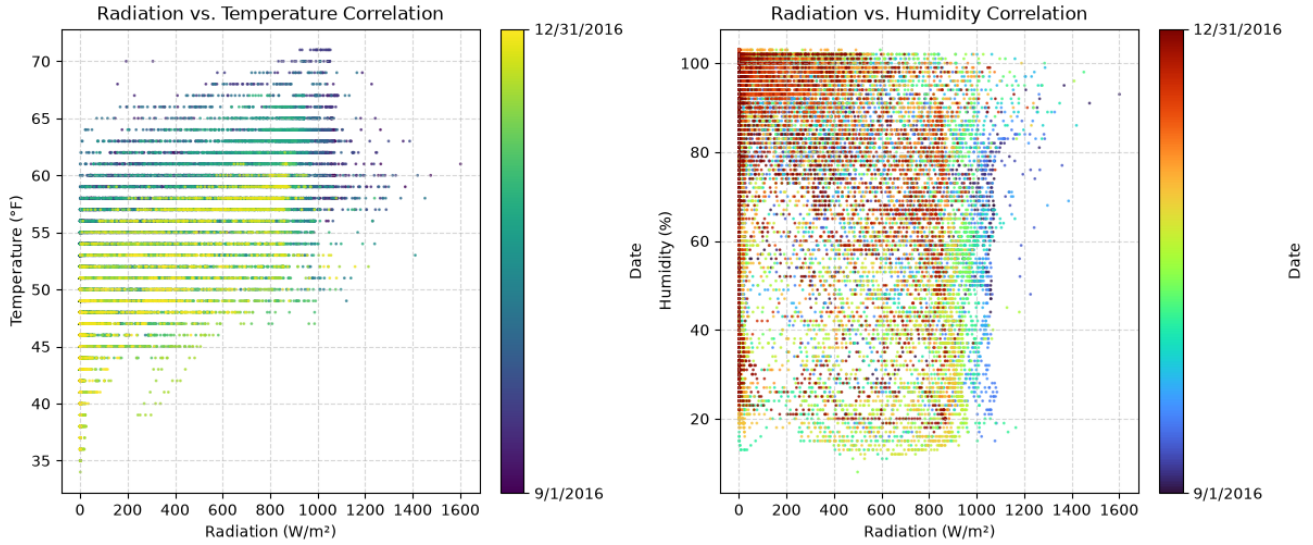


Fig. 2. Radiation against temperature (left) and against humidity (right), coloured by date.

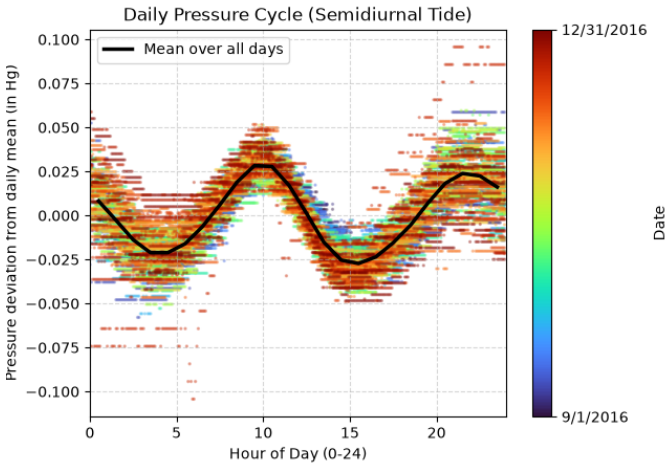


Fig. 3. Pressure deviation from the daily mean against hour of day, with the mean over all days in black.

IV. Part B: Drawing Inferences

A. When is radiation highest?

Mean radiation by hour rises from 128.8 W/m^2 at 07:00 to a maximum of 727.5 W/m^2 in the 12:00–13:00 hour, and falls to 54.6 W/m^2 at 17:00. Taking each day's single highest reading, the peak occurs on average at 12:14:47 and the median is 12:15:02. Radiation is therefore highest around 12:15:00 (solar noon).

B. Which parameter is most strongly related to radiation?

Two cautions apply. First, temperature correlates with radiation partly because both fall from September to December, so season inflates the figure. Second, time of day is the dominant driver, the relationship being a bell curve rather than a line. Linear correlation understates it, and Section VI confirms its importance with a non-linear model.

C. Radiation, temperature and humidity

Radiation and temperature move together: temperature rises as the sun climbs and the high-radiation envelope sits at the warm end of the temperature axis. Radiation and humidity are inversely related, and temperature and humidity are themselves negatively correlated. The pattern is consistent with warm, dry, clear conditions producing high radiation and cool, humid, overcast conditions suppressing it.

D. Outliers and unusual observations

Three features stand out.

1) *Cloud-affected scatter.* The mass of points far below the clear-sky envelope in Fig. 1 is attenuation by cloud. The extreme case is 9 September, with a daily mean of only 19.1 W/m^2, followed by 8 September (25.1) and 18 December (27.2).
2) *Readings above the envelope.* 49 readings exceed 1200 W/m^2 and 5 exceed 1400 W/m^2, while the median daily maximum in September is 1088 W/m^2. Brief over-shoots of this kind are consistent with cloud-edge enhancement, where sunlight scattered from the edges of clouds adds to the direct beam.
3) *The global maximum.* The highest value, 1601.26 W/m^2, exceeds the solar constant of about 1361 W/m^2 [**?**]. It was recorded at 93% humidity, an overcast-type condition, so it is best treated as a transient enhancement or a sensor spike rather than representative irradiance.

E. Bonus insight

The daily pressure cycle in Fig. 3 is a regular semidiurnal oscillation of about ± 0.025 inHg. It is small compared with weather-driven pressure changes, which is why it only appears once each day's mean is removed. Its timing is tied to

TABLE I
FIVE DAYS WITH THE HIGHEST AVERAGE RADIATION

Rank	Date	Mean radiation (W/m^2)
1	19 September 2016	335.50
2	2 September 2016	333.47
3	22 September 2016	325.52
4	25 September 2016	319.92
5	28 September 2016	312.40

the clock rather than to the sun's elevation, and pressure's own correlation with radiation is weak (0.119), so it is a daily rhythm of the atmosphere rather than a useful radiation predictor.

V. Part C: Solar Day Analysis

Date and time were merged into one datetime column, observations were grouped by calendar date, and the daily maximum and mean radiation were computed, giving a report with one row for each of the 118 days.

Highest radiation. The maximum, 1601.26 W/m^2, was recorded on 4 September 2016 at 12:15:04. At that moment the temperature was 61°F, the humidity 93%, and the wind speed 9.0 (dataset units).

Peak relative to sunrise and sunset. On that day the sun rose at 06:08:00 and set at 18:35:00, a daylight period of 12 h 27 min. The peak occurred 6 h 07 min 04 s after sunrise and 6 h 19 min 56 s before sunset. The midpoint of daylight is 12:21:30, so the peak fell about 6.5 minutes before solar noon.

Top five days by average radiation. Table I lists them. All five are in September, which matches the seasonal decline seen in Part A.

VI. Bonus: Solar Radiation Prediction

A. Features and model

The target is radiation. The inputs are temperature, humidity, pressure, wind speed, wind direction, hour, and minutes since midnight. A random forest regressor [**?**] with 100 trees and a fixed random seed (42) was used, because it handles non-linear and interacting relationships without feature scaling. The data were split randomly into 70% training (22 880 records) and 30% testing (9 806 records).

B. Scores

Table II reports the test scores. On the random split the model reaches an RMSE of 91.45 W/m^2 and $R^2 = 0.915$, so it explains about 91.5% of the variance in radiation. Fig. 4 compares predicted and actual values and shows the points tight around the ideal line near zero and widening through the mid-to-high range, where cloud cover changes radiation faster than the available inputs can explain.

A random split is optimistic for time-series data, because readings a few minutes apart land in both training and test sets. As a stricter check, the model was retrained on the first 70% of days and tested on the remaining days (from 23 November). The score drops to an RMSE of 178.42 W/m^2 and $R^2 = 0.526$. Part of this drop is a seasonal shift: the test period contains the lowest sun and coolest temperatures, and a forest cannot extrapolate beyond the range it was trained on. The chronological figure is the more honest estimate of forecasting skill.

TABLE II
RANDOM FOREST TEST SCORES

Split	RMSE (W/m^2)	R^2
Random 70/30 (seed 42)	91.45	0.915
Chronological 70/30	178.42	0.526

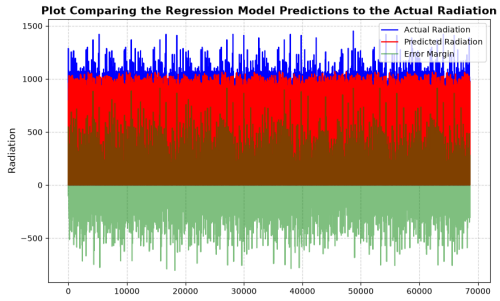


Fig. 4. Predicted against actual radiation on the random 30% test set.

VII. Conclusion

Radiation follows a regular daily bell curve centred on solar noon, with its ceiling falling about 21% from September to December and cloud producing most of the scatter beneath it. Temperature and time of day are the dominant predictors; humidity, pressure and wind add little. The day with the highest recorded radiation, 4 September, produced a value above the solar constant and should be treated as a transient or sensor artefact. A random forest scores $R^2 = 0.915$ on a random split but only 0.526 on a chronological split, so any forecasting claim should rest on the latter. A natural next step is to add features that carry the clear-sky expectation, such as time relative to sunrise and sunset and day of year, and to validate across a full year of data.

Code Availability

The Python scripts for Parts A–D are in the project GitHub repository: https://github.com/aarr699/STACWARS_Team-Nexus.